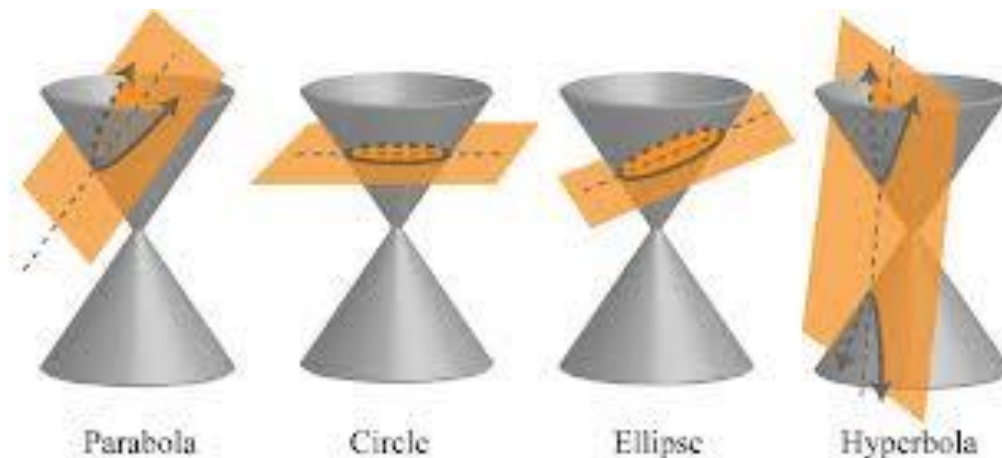


5.4.2 Conic Sections (Ellipses)

If a plane passes through a double right circular cone there are four possible resulting curves. These curves are the **parabola**, **circle**, **ellipse**, and **hyperbola**. These curves are called **conic sections** and are shown below.



Definition: An **ellipse** is the set of all points in a plane such that the sum of their distances from two fixed points is constant. Each fixed point is called a **focus** (plural **foci**) of the ellipse.

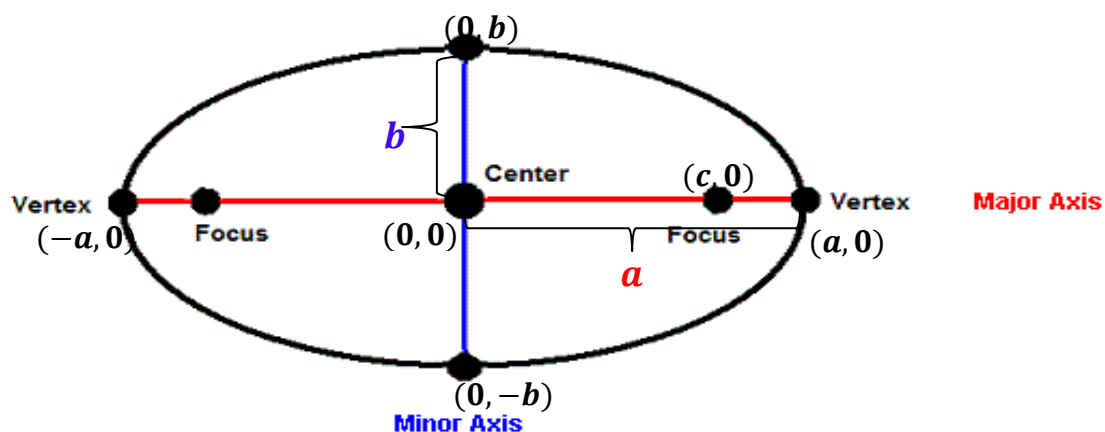
The Equation of an Ellipse: (with a *horizontal* major axis and center at the origin)

The ellipse with center at the origin and equation

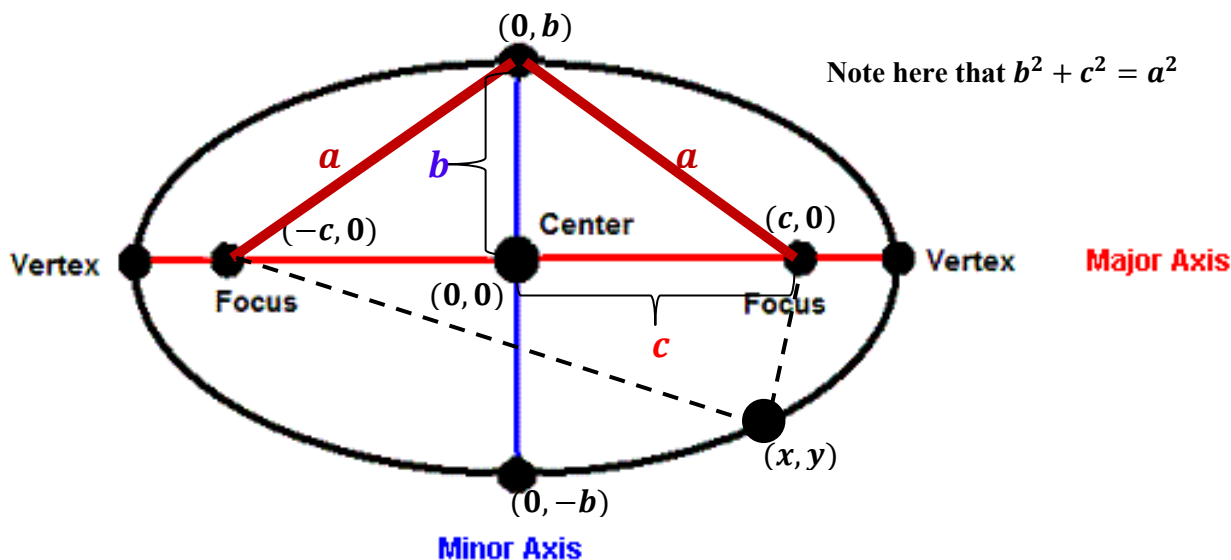
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b$$

has vertices $(\pm a, 0)$, endpoints of the minor axis $(0, \pm b)$, and foci $(\pm c, 0)$.

where $c^2 = a^2 - b^2$



Developing the Equation of an Ellipse: (Not done in class)



We see that the sum of the distances from $(-c, 0)$ to (x, y) and from $(c, 0)$ to (x, y) should be equal to $2a$.

$$\begin{aligned}
 & \sqrt{(x+c)^2 + (y-0)^2} + \sqrt{(x-c)^2 + (y-0)^2} = 2a \\
 \Leftrightarrow & \sqrt{(x+c)^2 + y^2} = 2a - \sqrt{(x-c)^2 + y^2} \\
 \Leftrightarrow & (x+c)^2 + y^2 = 4a^2 - 4a\sqrt{(x-c)^2 + y^2} + (x-c)^2 + y^2 \\
 \Leftrightarrow & x^2 + 2cx + c^2 + y^2 = 4a^2 - 4a\sqrt{(x-c)^2 + y^2} + x^2 - 2cx + c^2 + y^2 \\
 \Leftrightarrow & 4cx - 4a^2 = 4a\sqrt{(x-c)^2 + y^2} \\
 \Leftrightarrow & cx - a^2 = a\sqrt{(x-c)^2 + y^2} \\
 \Leftrightarrow & c^2x^2 - 2a^2cx + a^4 = a^2[(x-c)^2 + y^2] \\
 \Leftrightarrow & c^2x^2 - 2a^2cx + a^4 = a^2[x^2 - 2cx + c^2 + y^2] \\
 \Leftrightarrow & c^2x^2 - 2a^2cx + a^4 = a^2x^2 - 2a^2cx + a^2c^2 + a^2y^2 \\
 \Leftrightarrow & c^2x^2 + a^4 = a^2x^2 + a^2c^2 + a^2y^2 \\
 \Leftrightarrow & a^4 - a^2c^2 = a^2x^2 - c^2x^2 + a^2y^2 \\
 \Leftrightarrow & a^2(a^2 - c^2) = x^2(a^2 - c^2) + a^2y^2 \\
 \Leftrightarrow & a^2(a^2 - c^2) = x^2(a^2 - c^2) + a^2y^2
 \end{aligned}$$

Since we can see from our figure that $b^2 + c^2 = a^2$ we then know that $b^2 = a^2 - c^2$. So we have

$$a^2(a^2 - c^2) = x^2(a^2 - c^2) + a^2y^2 \Leftrightarrow a^2b^2 = x^2b^2 + a^2y^2 \Leftrightarrow 1 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

Also note that if let $y=0$ then we can find our x-intercepts. $\frac{x^2}{a^2} = \frac{(0)^2}{b^2} = 1 \Leftrightarrow x = \pm a$

So we have x-intercepts at $(\pm a, 0)$

The Equation of an Ellipse: (with a vertical major axis and center at the origin)

The ellipse with center at the origin and equation

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1, \quad a > b$$

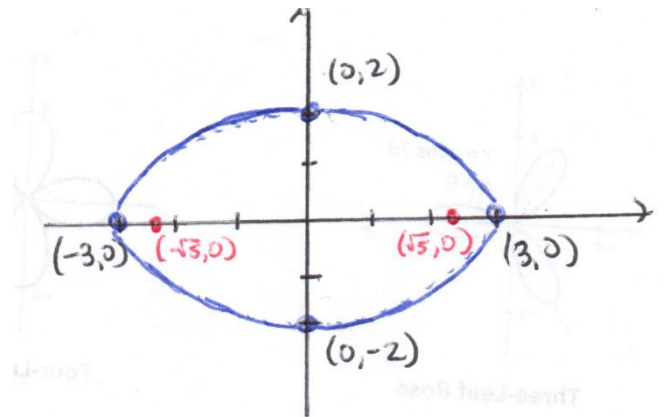
has vertices $(0, \pm a)$, endpoints of the minor axis $(\pm b, 0)$, and foci $(0, \pm c)$.

where $c^2 = a^2 - b^2$

Example 1: Sketch a graph and identify the foci of the ellipse:

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$a^2 = 9, \quad b^2 = 4 \quad c^2 = 9 - 4 = 5, \quad c = \pm\sqrt{5}$$



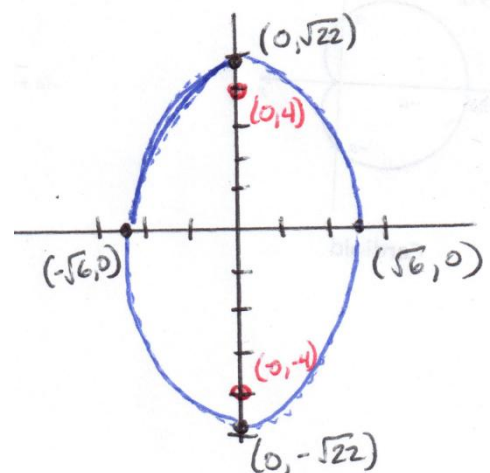
Example 2: Sketch a graph and identify the foci of the ellipse:

$$11x^2 + 3y^2 = 66$$

We first put this ellipse in standard form.

$$\frac{x^2}{6} + \frac{y^2}{22} = 1$$

$$a^2 = 22, \quad b^2 = 6 \quad c^2 = 22 - 6 = 16, \quad c = \pm 4$$



Example 3: Write the equation of an ellipse with foci $(0, \pm 4)$ and vertices $(0, \pm 6)$.

Since our foci are given by the points $(0, \pm c)$ we see that

$$c = \pm 4 \Rightarrow c^2 = 16$$

Since our vertices are given by the points $(0, \pm a)$ we see that

$$a = \pm 6 \Rightarrow a^2 = 36$$

We can also see that since our vertices lie on the y -axis, our ellipse has a vertical major axis.

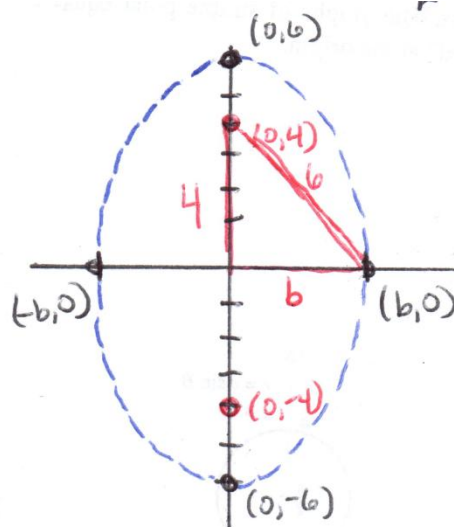
$$c^2 = a^2 - b^2$$

$$16 = 36 - b^2$$

$$b^2 = 20$$

Therefore the equation of our ellipse is:

$$\frac{x^2}{20} + \frac{y^2}{36} = 1$$



Fact: The equation representing a conic that has been shifted h units horizontally and k units vertically can be formed by replacing x 's in the original equation with $x - h$ and replacing any y 's in the original equation with $y - k$.

Example 4: Sketch a graph and identify the foci of the ellipse: $9x^2 - 54x - 19 = 100 - 50y - 25y^2$

We must use the method of completing the square to get the ellipse into standard form

$$9x^2 - 54x - 19 = 100 - 50y - 25y^2 \Leftrightarrow 9x^2 - 54x + 25y^2 + 50y = 119 \Leftrightarrow 9(x^2 - 6x + \quad) + 25(y^2 + 2y + \quad) = 119$$

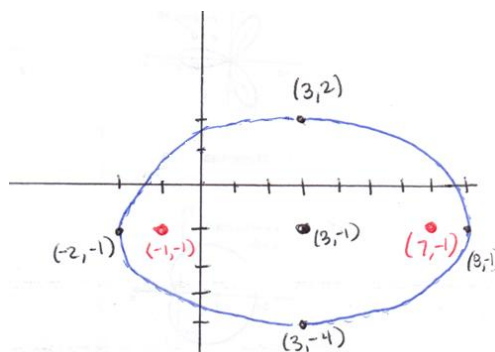
$$\Leftrightarrow 9(x^2 - 6x + 9) + 25(y^2 + 2y + 1) = 119 + 81 + 25$$

$$\Leftrightarrow 9(x - 3)^2 + 25(y + 1)^2 = 225$$

$$\Leftrightarrow \frac{(x - 3)^2}{25} + \frac{(y + 1)^2}{9} = 1$$

$$a^2 = 25, \quad b^2 = 9 \quad c^2 = 25 - 9 = 16, \quad c = \pm 4$$

The center of the ellipse occurs at the point: $(3, -1)$



The Circle

Definition: A circle is a set of all points in a plane such that their distance from a point is constant.

Equation for a Circle

The standard equation of a circle with center (h, k) and radius r ($r > 0$) is

$$(x - h)^2 + (y - k)^2 = r^2$$

Note: By dividing both sides of this equation by r^2 we obtain $\frac{(x-h)^2}{r^2} + \frac{(y-k)^2}{r^2} = 1$. So clearly a circle is an ellipse with major and minor axes of the same length.

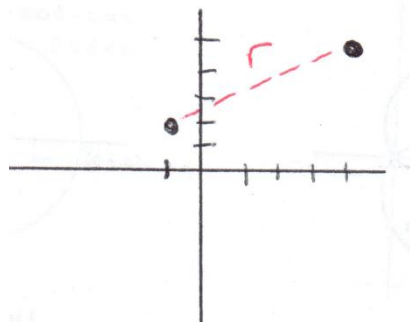
Example 5: Write the equation of a circle that has center $(4, 5)$ and passes through the point $(-1, 2)$.

$$r^2 = (-1 - 4)^2 + (2 - 5)^2$$

$$r^2 = 25 + 9 = 34$$

So the equation of the circle is:

$$(x - 4)^2 + (y - 5)^2 = 34$$



Example 6: Determine the center and radius and sketch a graph of the circle given by the equation:

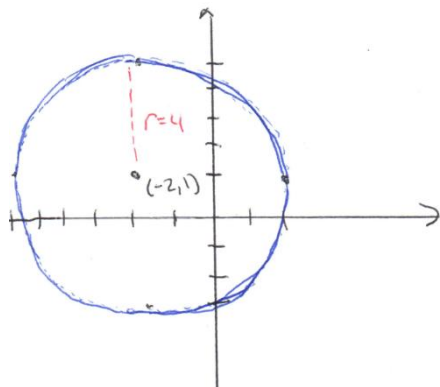
$$x^2 + 4x + y^2 - 2y = 11$$

We must use the method of completing the square to get the circle into standard form.

$$(x^2 + 4x + \underline{\quad}) + (y^2 - 2y + \underline{\quad}) = 11$$

$$\Rightarrow (x^2 + 4x + 4) + (y^2 - 2y + 1) = 11 + 4 + 1$$

$$\Leftrightarrow (x + 2)^2 + (y - 1)^2 = 16$$



So we have a circle with center $(-2, 1)$ and radius $r = \sqrt{16} = 4$.